Instrumenting Python code in AWS Lambda

Lambda integrates with AWS X-Ray to help you trace, debug, and optimize Lambda applications. You can use X-Ray to trace a request as it traverses resources in your application, which may include Lambda functions and other AWS services.

To send tracing data to X-Ray, you can use one of three SDK libraries:

- AWS Distro for OpenTelemetry (ADOT) – A secure, production-ready, AWS-supported distribution of the OpenTelemetry (OTel) SDK.
- AWS X-Ray SDK for Python – An SDK for generating and sending trace data to X-Ray.
- Powertools for AWS Lambda (Python) – A developer toolkit to implement Serverless best practices and increase developer velocity.

Each of the SDKs offer ways to send your telemetry data to the X-Ray service. You can then use X-Ray to view, filter, and gain insights into your application's performance metrics to identify issues and opportunities for optimization.

> **Important**
>
> The X-Ray and Powertools for AWS Lambda SDKs are part of a tightly integrated instrumentation solution offered by AWS. The ADOT Lambda Layers are part of an industry-wide standard for tracing instrumentation that collect more data in general, but may not be suited for all use cases. You can implement end-to-end tracing in X-Ray using either solution. To learn more about choosing between them, see Choosing between the AWS Distro for Open Telemetry and X-Ray SDKs.

Sections

- Using Powertools for AWS Lambda (Python) and AWS SAM for tracing
- Using Powertools for AWS Lambda (Python) and the AWS CDK for tracing
- Using ADOT to instrument your Python functions
- Using the X-Ray SDK to instrument your Python functions
- Activating tracing with the Lambda console
- Activating tracing with the Lambda API
- Activating tracing with CloudFormation

- Interpreting an X-Ray trace
- Storing runtime dependencies in a layer (X-Ray SDK)

Using Powertools for AWS Lambda (Python) and AWS SAM for tracing

Follow the steps below to download, build, and deploy a sample Hello World Python application with integrated Powertools for AWS Lambda (Python) modules using the AWS SAM. This application implements a basic API backend and uses Powertools for emitting logs, metrics, and traces. It consists of an Amazon API Gateway endpoint and a Lambda function. When you send a GET request to the API Gateway endpoint, the Lambda function invokes, sends logs and metrics using Embedded Metric Format to CloudWatch, and sends traces to AWS X-Ray. The function returns a hello world message.

Prerequisites

To complete the steps in this section, you must have the following:

- Python 3.11
- AWS CLI version 2
- AWS SAM CLI version 1.75 or later. If you have an older version of the AWS SAM CLI, see Upgrading the AWS SAM CLI.

Deploy a sample AWS SAM application

1. Initialize the application using the Hello World Python template.

```
sam init --app-template hello-world-powertools-python --name sam-app --package-type
 Zip --runtime python3.11 --no-tracing
```

2. Build the app.

```
cd sam-app && sam build
```

3. Deploy the app.

```
sam deploy --guided
```

4. Follow the on-screen prompts. To accept the default options provided in the interactive experience, press `Enter`.

> **Note**
>
> For **HelloWorldFunction may not have authorization defined, Is this okay?**, make sure to enter y.

5. Get the URL of the deployed application:

```
aws cloudformation describe-stacks --stack-name sam-app --query
 'Stacks[0].Outputs[?OutputKey==`HelloWorldApi`].OutputValue' --output text
```

6. Invoke the API endpoint:

```
curl -X GET <URL_FROM_PREVIOUS_STEP>
```

If successful, you'll see this response:

```
{"message":"hello world"}
```

7. To get the traces for the function, run sam traces.

```
sam traces
```

The trace output looks like this:

```
New XRay Service Graph
  Start time: 2023-02-03 14:59:50+00:00
  End time: 2023-02-03 14:59:50+00:00
  Reference Id: 0 - (Root) AWS::Lambda - sam-app-HelloWorldFunction-YBg8yfYtOc9j -
 Edges: [1]
   Summary_statistics:
     - total requests: 1
     - ok count(2XX): 1
     - error count(4XX): 0
     - fault count(5XX): 0
     - total response time: 0.924
  Reference Id: 1 - AWS::Lambda::Function - sam-app-HelloWorldFunction-YBg8yfYtOc9j
 - Edges: []
   Summary_statistics:
     - total requests: 1
     - ok count(2XX): 1
```

```
      - error count(4XX): 0
      - fault count(5XX): 0
      - total response time: 0.016
  Reference Id: 2 - client - sam-app-HelloWorldFunction-YBg8yfYtOc9j - Edges: [0]
   Summary_statistics:
      - total requests: 0
      - ok count(2XX): 0
      - error count(4XX): 0
      - fault count(5XX): 0
      - total response time: 0

XRay Event [revision 1] at (2023-02-03T14:59:50.204000) with id
 (1-63dd2166-434a12c22e1307ff2114f299) and duration (0.924s)
 - 0.924s - sam-app-HelloWorldFunction-YBg8yfYtOc9j [HTTP: 200]
 - 0.016s - sam-app-HelloWorldFunction-YBg8yfYtOc9j
   - 0.739s - Initialization
   - 0.016s - Invocation
     - 0.013s - ## lambda_handler
       - 0.000s - ## app.hello
   - 0.000s - Overhead
```

8. This is a public API endpoint that is accessible over the internet. We recommend that you delete the endpoint after testing.

```
sam delete
```

X-Ray doesn't trace all requests to your application. X-Ray applies a sampling algorithm to ensure that tracing is efficient, while still providing a representative sample of all requests. The sampling rate is 1 request per second and 5 percent of additional requests. You can't configure the X-Ray sampling rate for your functions.

Using Powertools for AWS Lambda (Python) and the AWS CDK for tracing

Follow the steps below to download, build, and deploy a sample Hello World Python application with integrated Powertools for AWS Lambda (Python) modules using the AWS CDK. This application implements a basic API backend and uses Powertools for emitting logs, metrics, and traces. It consists of an Amazon API Gateway endpoint and a Lambda function. When you send a GET request to the API Gateway endpoint, the Lambda function invokes, sends logs and metrics

using Embedded Metric Format to CloudWatch, and sends traces to AWS X-Ray. The function returns a hello world message.

Prerequisites

To complete the steps in this section, you must have the following:

- Python 3.11
- AWS CLI version 2
- AWS CDK version 2
- AWS SAM CLI version 1.75 or later. If you have an older version of the AWS SAM CLI, see Upgrading the AWS SAM CLI.

Deploy a sample AWS CDK application

1. Create a project directory for your new application.

```
mkdir hello-world
cd hello-world
```

2. Initialize the app.

```
cdk init app --language python
```

3. Install the Python dependencies.

```
pip install -r requirements.txt
```

4. Create a directory **lambda_function** under the root folder.

```
mkdir lambda_function
cd lambda_function
```

5. Create a file **app.py** and add the following code to the file. This is the code for the Lambda function.

```
from aws_lambda_powertools.event_handler import APIGatewayRestResolver
from aws_lambda_powertools.utilities.typing import LambdaContext
from aws_lambda_powertools.logging import correlation_paths
from aws_lambda_powertools import Logger
```

```
from aws_lambda_powertools import Tracer
from aws_lambda_powertools import Metrics
from aws_lambda_powertools.metrics import MetricUnit

app = APIGatewayRestResolver()
tracer = Tracer()
logger = Logger()
metrics = Metrics(namespace="PowertoolsSample")

@app.get("/hello")
@tracer.capture_method
def hello():
    # adding custom metrics
    # See: https://docs.powertools.aws.dev/lambda-python/latest/core/metrics/
    metrics.add_metric(name="HelloWorldInvocations", unit=MetricUnit.Count,
 value=1)

    # structured log
    # See: https://docs.powertools.aws.dev/lambda-python/latest/core/logger/
    logger.info("Hello world API - HTTP 200")
    return {"message": "hello world"}

# Enrich logging with contextual information from Lambda
@logger.inject_lambda_context(correlation_id_path=correlation_paths.API_GATEWAY_REST)
# Adding tracer
# See: https://docs.powertools.aws.dev/lambda-python/latest/core/tracer/
@tracer.capture_lambda_handler
# ensures metrics are flushed upon request completion/failure and capturing
 ColdStart metric
@metrics.log_metrics(capture_cold_start_metric=True)
def lambda_handler(event: dict, context: LambdaContext) -> dict:
    return app.resolve(event, context)
```

6. Open the **hello_world** directory. You should see a file called **hello_world_stack.py**.

```
cd ..
cd hello_world
```

7. Open **hello_world_stack.py** and add the following code to the file. This contains the Lambda Constructor, which creates the Lambda function, configures environment variables for Powertools and sets log retention to one week, and the ApiGatewayv1 Constructor, which creates the REST API.

```
from aws_cdk import (
    Stack,
    aws_apigateway as apigwv1,
    aws_lambda as lambda_,
    CfnOutput,
    Duration
)
from constructs import Construct

class HelloWorldStack(Stack):

    def __init__(self, scope: Construct, construct_id: str, **kwargs) -> None:
        super().__init__(scope, construct_id, **kwargs)

        # Powertools Lambda Layer
        powertools_layer = lambda_.LayerVersion.from_layer_version_arn(
            self,
            id="lambda-powertools",
            # At the moment we wrote this example, the aws_lambda_python_alpha CDK
 constructor is in Alpha, o we use layer to make the example simpler
            # See https://docs.aws.amazon.com/cdk/api/v2/python/
aws_cdk.aws_lambda_python_alpha/README.html
            # Check all Powertools layers versions here: https://
docs.powertools.aws.dev/lambda-python/latest/#lambda-layer
            layer_version_arn=f"arn:aws:lambda:
{self.region}:017000801446:layer:AWSLambdaPowertoolsPythonV2:21"
        )

        function = lambda_.Function(self,
            'sample-app-lambda',
            runtime=lambda_.Runtime.PYTHON_3_11,
            layers=[powertools_layer],
            code = lambda_.Code.from_asset("./lambda_function/"),
            handler="app.lambda_handler",
            memory_size=128,
            timeout=Duration.seconds(3),
            architecture=lambda_.Architecture.X86_64,
            environment={
                "POWERTOOLS_SERVICE_NAME": "PowertoolsHelloWorld",
                "POWERTOOLS_METRICS_NAMESPACE": "PowertoolsSample",
                "LOG_LEVEL": "INFO"
            }
        )
```

```
        apigw = apigwv1.RestApi(self, "PowertoolsAPI",
 deploy_options=apigwv1.StageOptions(stage_name="dev"))

        hello_api = apigw.root.add_resource("hello")
        hello_api.add_method("GET", apigwv1.LambdaIntegration(function,
 proxy=True))

        CfnOutput(self, "apiUrl", value=f"{apigw.url}hello")
```

8. Deploy your application.

```
cd ..
cdk deploy
```

9. Get the URL of the deployed application:

```
aws cloudformation describe-stacks --stack-name HelloWorldStack --query
 'Stacks[0].Outputs[?OutputKey==`apiUrl`].OutputValue' --output text
```

10. Invoke the API endpoint:

```
curl -X GET <URL_FROM_PREVIOUS_STEP>
```

If successful, you'll see this response:

```
{"message":"hello world"}
```

11. To get the traces for the function, run sam traces.

```
sam traces
```

The traces output looks like this:

```
New XRay Service Graph
  Start time: 2023-02-03 14:59:50+00:00
  End time: 2023-02-03 14:59:50+00:00
  Reference Id: 0 - (Root) AWS::Lambda - sam-app-HelloWorldFunction-YBg8yfYtOc9j -
 Edges: [1]
   Summary_statistics:
     - total requests: 1
     - ok count(2XX): 1
```

```
      - error count(4XX): 0
      - fault count(5XX): 0
      - total response time: 0.924
  Reference Id: 1 - AWS::Lambda::Function - sam-app-HelloWorldFunction-YBg8yfYtOc9j
 - Edges: []
   Summary_statistics:
      - total requests: 1
      - ok count(2XX): 1
      - error count(4XX): 0
      - fault count(5XX): 0
      - total response time: 0.016
  Reference Id: 2 - client - sam-app-HelloWorldFunction-YBg8yfYtOc9j - Edges: [0]
   Summary_statistics:
      - total requests: 0
      - ok count(2XX): 0
      - error count(4XX): 0
      - fault count(5XX): 0
      - total response time: 0

XRay Event [revision 1] at (2023-02-03T14:59:50.204000) with id
 (1-63dd2166-434a12c22e1307ff2114f299) and duration (0.924s)
 - 0.924s - sam-app-HelloWorldFunction-YBg8yfYtOc9j [HTTP: 200]
 - 0.016s - sam-app-HelloWorldFunction-YBg8yfYtOc9j
   - 0.739s - Initialization
   - 0.016s - Invocation
     - 0.013s - ## lambda_handler
       - 0.000s - ## app.hello
   - 0.000s - Overhead
```

12. This is a public API endpoint that is accessible over the internet. We recommend that you delete the endpoint after testing.

```
cdk destroy
```

Using ADOT to instrument your Python functions

ADOT provides fully managed Lambda layers that package everything you need to collect telemetry data using the OTel SDK. By consuming this layer, you can instrument your Lambda functions without having to modify any function code. You can also configure your layer to do custom initialization of OTel. For more information, see Custom configuration for the ADOT Collector on Lambda in the ADOT documentation.

For Python runtimes, you can add the **AWS managed Lambda layer for ADOT Python** to automatically instrument your functions. This layer works for both arm64 and x86_64 architectures. For detailed instructions on how to add this layer, see AWS Distro for OpenTelemetry Lambda Support for Python in the ADOT documentation.

Using the X-Ray SDK to instrument your Python functions

To record details about calls that your Lambda function makes to other resources in your application, you can also use the AWS X-Ray SDK for Python. To get the SDK, add the `aws-xray-sdk` package to your application's dependencies.

Example requirements.txt

```
jsonpickle==1.3
aws-xray-sdk==2.4.3
```

In your function code, you can instrument AWS SDK clients by patching the `boto3` library with the `aws_xray_sdk.core` module.

Example function – Tracing an AWS SDK client

```
import boto3
from aws_xray_sdk.core import xray_recorder
from aws_xray_sdk.core import patch_all

logger = logging.getLogger()
logger.setLevel(logging.INFO)
patch_all()

client = boto3.client('lambda')
client.get_account_settings()

def lambda_handler(event, context):
    logger.info('## ENVIRONMENT VARIABLES\r' + jsonpickle.encode(dict(**os.environ)))
  ...
```

After you add the correct dependencies and make the necessary code changes, activate tracing in your function's configuration via the Lambda console or the API.

Activating tracing with the Lambda console

To toggle active tracing on your Lambda function with the console, follow these steps:

To turn on active tracing

1. Open the Functions page of the Lambda console.
2. Choose a function.
3. Choose **Configuration** and then choose **Monitoring and operations tools**.
4. Under **Additional monitoring tools**, choose **Edit**.
5. Under **CloudWatch Application Signals and AWS X-Ray**, choose **Enable** for **Lambda service traces**.
6. Choose **Save**.

Activating tracing with the Lambda API

Configure tracing on your Lambda function with the AWS CLI or AWS SDK, use the following API operations:

- UpdateFunctionConfiguration
- GetFunctionConfiguration
- CreateFunction

The following example AWS CLI command enables active tracing on a function named **my-function**.

```
aws lambda update-function-configuration --function-name my-function \
--tracing-config Mode=Active
```

Tracing mode is part of the version-specific configuration when you publish a version of your function. You can't change the tracing mode on a published version.

Activating tracing with CloudFormation

To activate tracing on an `AWS::Lambda::Function` resource in an CloudFormation template, use the `TracingConfig` property.

Example function-inline.yml – Tracing configuration

```
Resources:
  function:
    Type: AWS::Lambda::Function
    Properties:
      TracingConfig:
        Mode: Active
      ...
```

For an AWS Serverless Application Model (AWS SAM) `AWS::Serverless::Function` resource, use the `Tracing` property.

Example template.yml – Tracing configuration

```
Resources:
  function:
    Type: AWS::Serverless::Function
    Properties:
      Tracing: Active
      ...
```

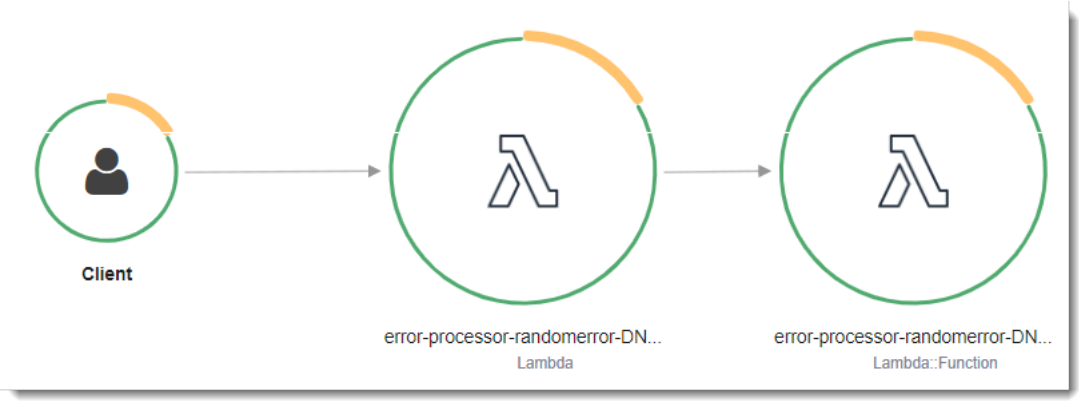
Interpreting an X-Ray trace

Your function needs permission to upload trace data to X-Ray. When you activate tracing in the Lambda console, Lambda adds the required permissions to your function's execution role. Otherwise, add the AWSXRayDaemonWriteAccess policy to the execution role.

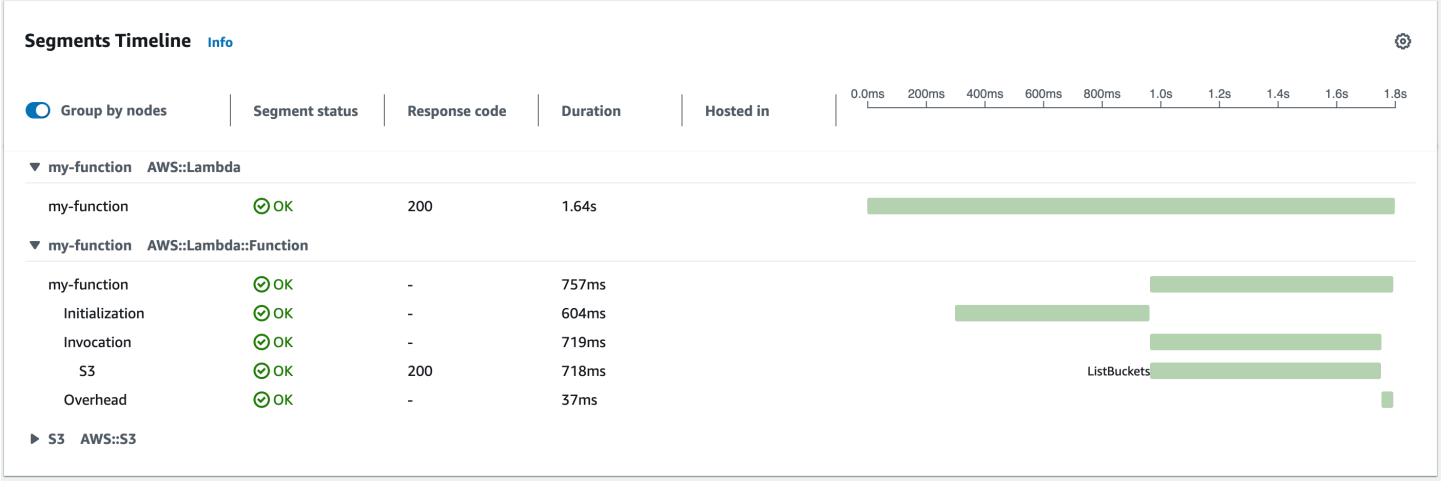
After you've configured active tracing, you can observe specific requests through your application. The X-Ray service graph shows information about your application and all its components. The following example shows an application with two functions. The primary function processes events and sometimes returns errors. The second function at the top processes errors that appear in the first's log group and uses the AWS SDK to call X-Ray, Amazon Simple Storage Service (Amazon S3), and Amazon CloudWatch Logs.

X-Ray doesn't trace all requests to your application. X-Ray applies a sampling algorithm to ensure that tracing is efficient, while still providing a representative sample of all requests. The sampling rate is 1 request per second and 5 percent of additional requests. You can't configure the X-Ray sampling rate for your functions.

In X-Ray, a *trace* records information about a request that is processed by one or more *services*. Lambda records 2 segments per trace, which creates two nodes on the service graph. The following image highlights these two nodes:



The first node on the left represents the Lambda service, which receives the invocation request. The second node represents your specific Lambda function. The following example shows a trace with these two segments. Both are named **my-function**, but one has an origin of `AWS::Lambda` and the other has an origin of `AWS::Lambda::Function`. If the `AWS::Lambda` segment shows an error, the Lambda service had an issue. If the `AWS::Lambda::Function` segment shows an error, your function had an issue.



This example expands the `AWS::Lambda::Function` segment to show its three subsegments.

> **Note**
>
> AWS is currently implementing changes to the Lambda service. Due to these changes, you may see minor differences between the structure and content of system log messages and trace segments emitted by different Lambda functions in your AWS account.
> The example trace shown here illustrates the old-style function segment. The differences between the old- and new-style segments are described in the following paragraphs.
> These changes will be implemented during the coming weeks, and all functions in all AWS Regions except the China and GovCloud regions will transition to use the new-format log messages and trace segments.

The old-style function segment contains the following subsegments:

- **Initialization** – Represents time spent loading your function and running initialization code. This subsegment only appears for the first event that each instance of your function processes.
- **Invocation** – Represents the time spent running your handler code.
- **Overhead** – Represents the time the Lambda runtime spends preparing to handle the next event.

The new-style function segment doesn't contain an `Invocation` subsegment. Instead, customer subsegments are attached directly to the function segment. For more information about the structure of the old- and new-style function segments, see the section called "Understanding X-Ray traces".

You can also instrument HTTP clients, record SQL queries, and create custom subsegments with annotations and metadata. For more information, see the AWS X-Ray SDK for Python in the *AWS X-Ray Developer Guide*.

> **Pricing**
>
> You can use X-Ray tracing for free each month up to a certain limit as part of the AWS Free Tier. Beyond that threshold, X-Ray charges for trace storage and retrieval. For more information, see AWS X-Ray pricing.

Storing runtime dependencies in a layer (X-Ray SDK)

If you use the X-Ray SDK to instrument AWS SDK clients your function code, your deployment package can become quite large. To avoid uploading runtime dependencies every time you update your function code, package the X-Ray SDK in a Lambda layer.

The following example shows an `AWS::Serverless::LayerVersion` resource that stores the AWS X-Ray SDK for Python.

Example template.yml – Dependencies layer

```
Resources:
  function:
    Type: AWS::Serverless::Function
    Properties:
      CodeUri: function/.
      Tracing: Active
      Layers:
        - !Ref libs
      ...
  libs:
    Type: AWS::Serverless::LayerVersion
    Properties:
      LayerName: blank-python-lib
      Description: Dependencies for the blank-python sample app.
      ContentUri: package/.
      CompatibleRuntimes:
        - python3.11
```

With this configuration, you update the library layer only if you change your runtime dependencies. Since the function deployment package contains only your code, this can help reduce upload times.

Creating a layer for dependencies requires build changes to generate the layer archive prior to deployment. For a working example, see the blank-python sample application.